Year 9

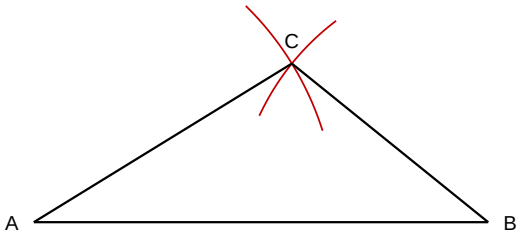
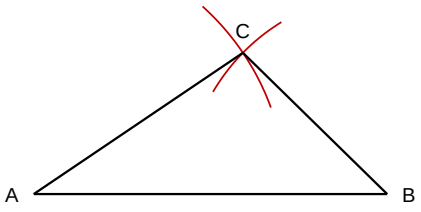
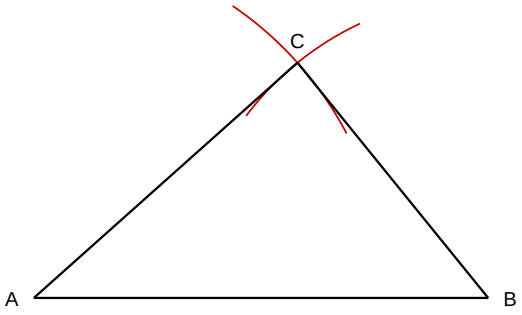
REVISION PACK

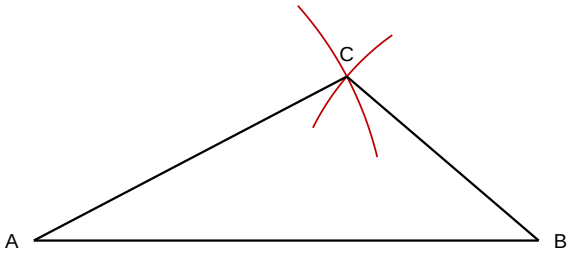
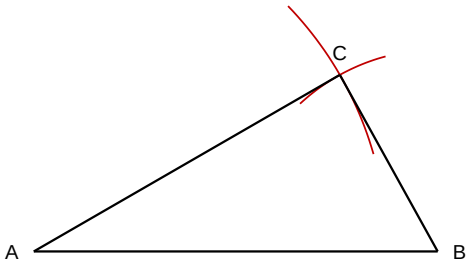
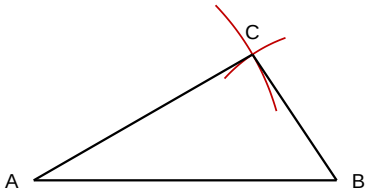
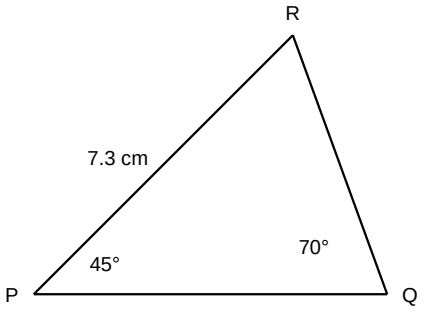
Unit 4

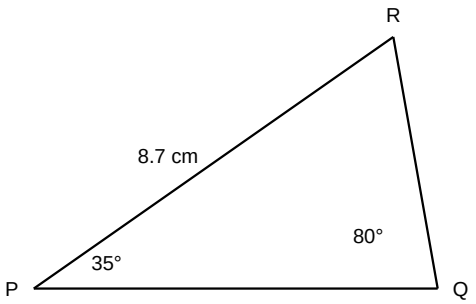
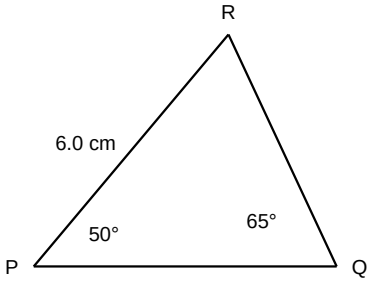
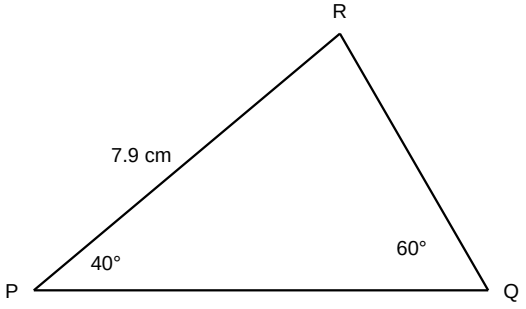
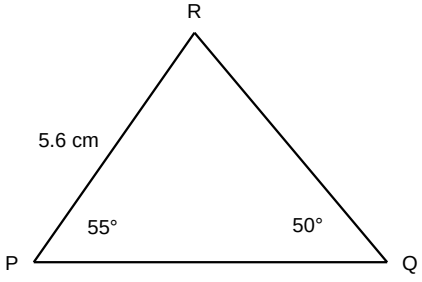
ANSWERS

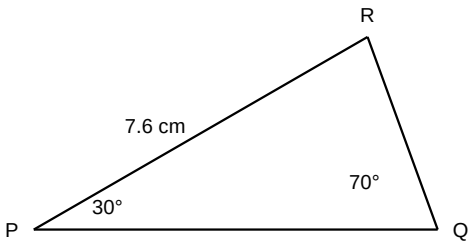
Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 4 · Constructing triangles (Lesson 4.1)

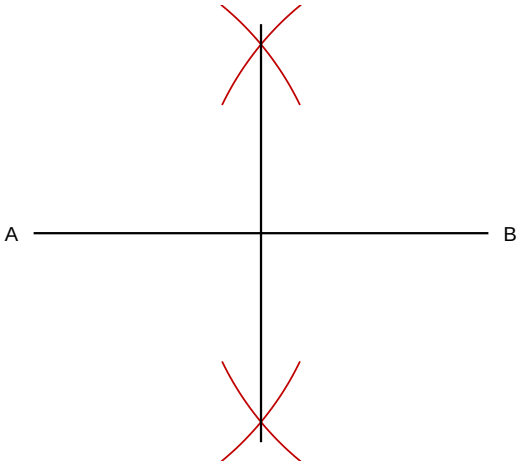
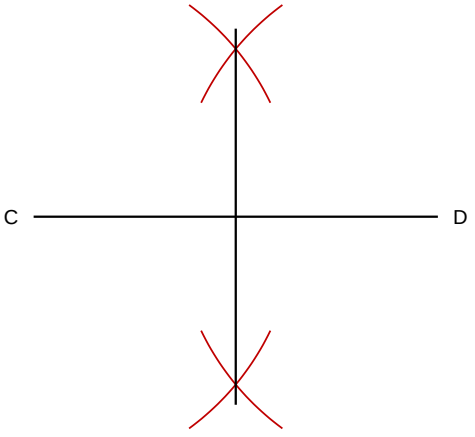
Q	Working	Answer
a triangle with three sides		
1.1		**Arcs of radius 6 cm from A and 5 cm from B meet at C**
1.2		**Arcs of radius 5 cm from A and 4 cm from B meet at C**
1.3		**Arcs of radius 7 cm from A and 6 cm from B meet at C**

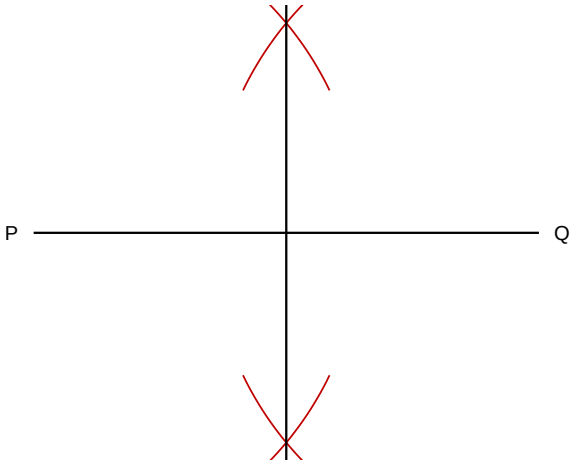
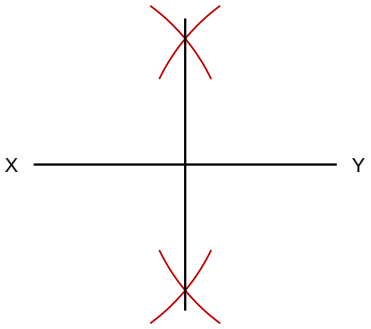
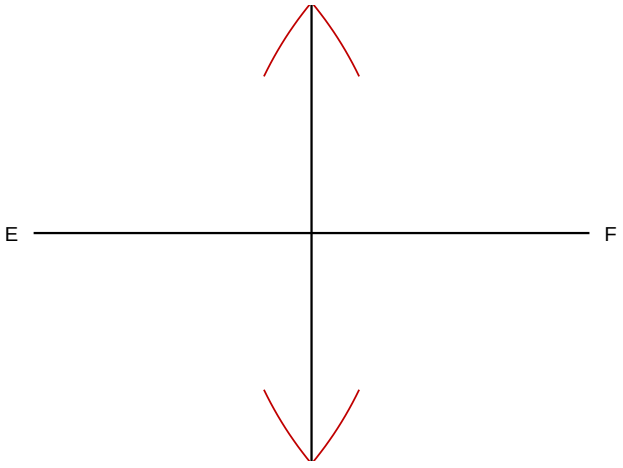
1.4		**Arcs of radius 7 cm from A and 5 cm from B meet at C**
1.5		**Arcs of radius 7 cm from A and 4 cm from B meet at C**
1.6		**Arcs of radius 5 cm from A and 3 cm from B meet at C**
a triangle with a side and two angles		
1.7		**PR = 7.3 cm (accept ±2 mm)**

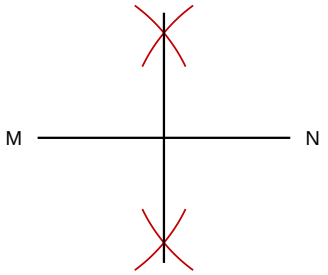
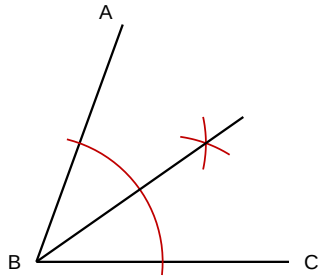
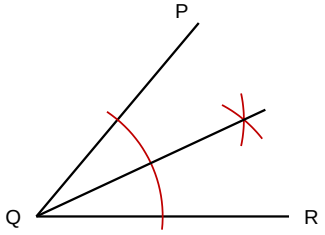
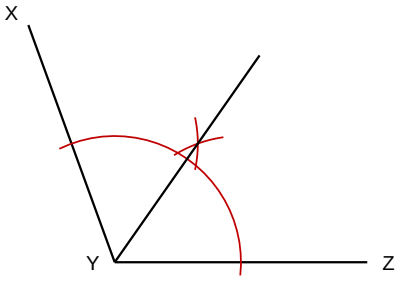
1.8		PR **= 8.7 cm (accept ±2 mm)**
1.9		PR **= 6.0 cm (accept ±2 mm)**
1.10		PR **= 7.9 cm (accept ±2 mm)**
1.11		PR **= 5.6 cm (accept ±2 mm)**

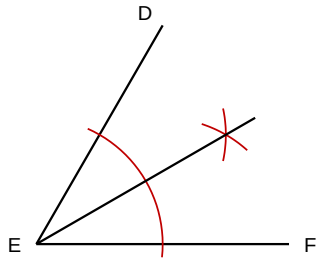
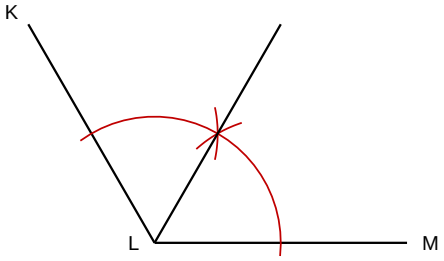
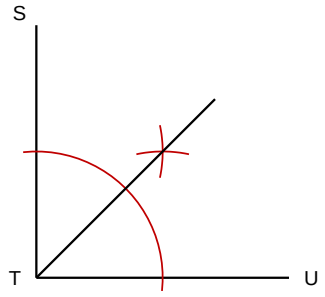
1.12		**PR = 7.6 cm (accept ±2 mm)**

Unit 4 · Bisectors (Lesson 4.2)

Q	Working	Answer
the perpendicular bisector of a line		
2.1		**Equal arcs from A and B above and below; the line crosses AB at 4.5 cm from each end at 90°**
2.2		**Equal arcs from C and D above and below; the line crosses CD at 4 cm from each end at 90°**

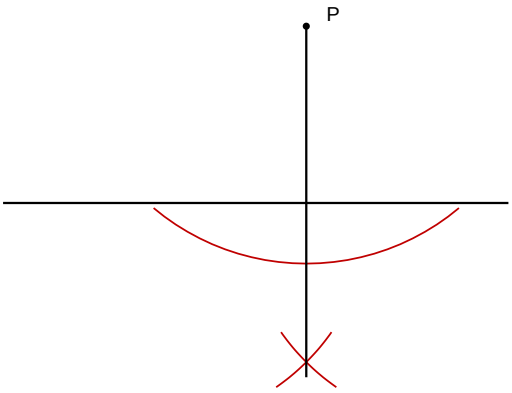
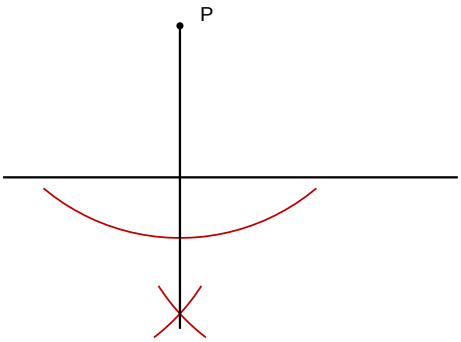
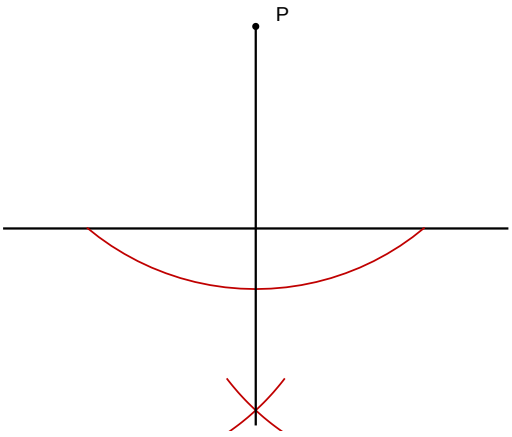
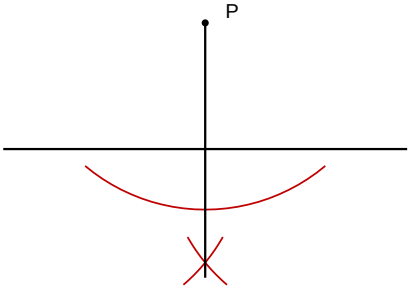
2.3		**Equal arcs from P and Q above and below; the line crosses PQ at 5 cm from each end at 90°**
2.4		**Equal arcs from X and Y above and below; the line crosses XY at 3 cm from each end at 90°**
2.5		**Equal arcs from E and F above and below; the line crosses EF at 5.5 cm from each end at 90°**

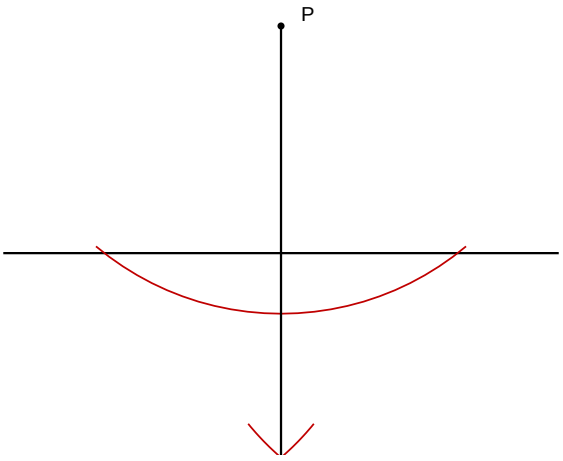
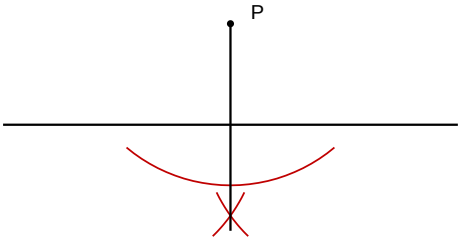
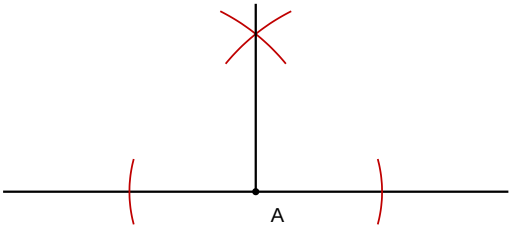
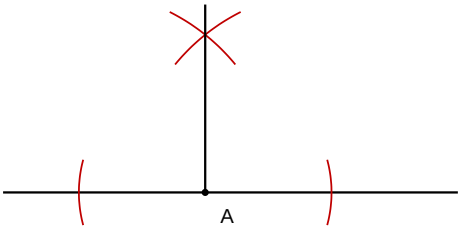
2.6		**Equal arcs from M and N above and below; the line crosses MN at 2.5 cm from each end at 90°**
the bisector of an angle		
2.7		**Arc from B cuts both arms; equal arcs from those points meet; the bisector makes 35° with each arm**
2.8		**Arc from Q cuts both arms; equal arcs from those points meet; the bisector makes 25° with each arm**
2.9		**Arc from Y cuts both arms; equal arcs from those points meet; the bisector makes 55° with each arm**

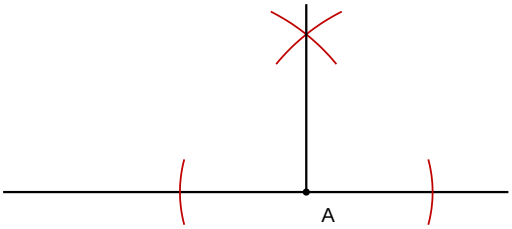
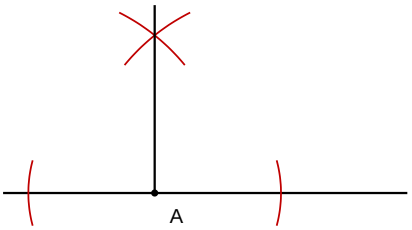
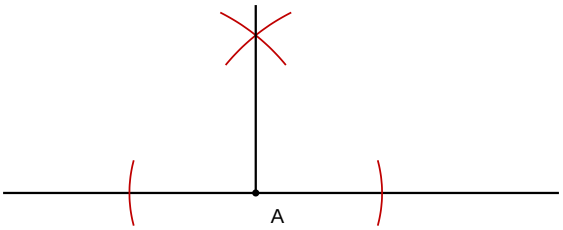
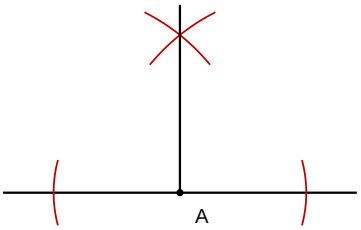
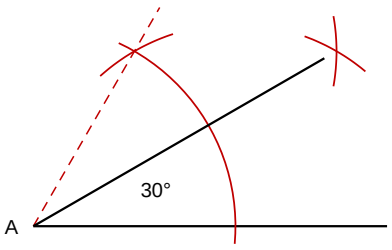
2.10		**Arc from E cuts both arms; equal arcs from those points meet; the bisector makes 30° with each arm**
2.11		**Arc from L cuts both arms; equal arcs from those points meet; the bisector makes 60° with each arm**
2.12		**Arc from T cuts both arms; equal arcs from those points meet; the bisector makes 45° with each arm**

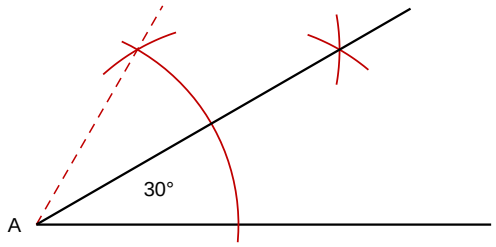
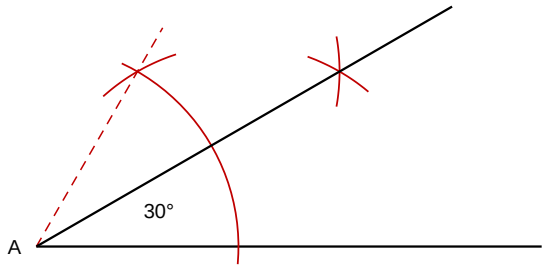
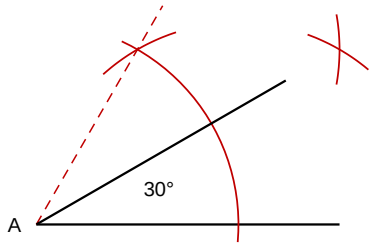
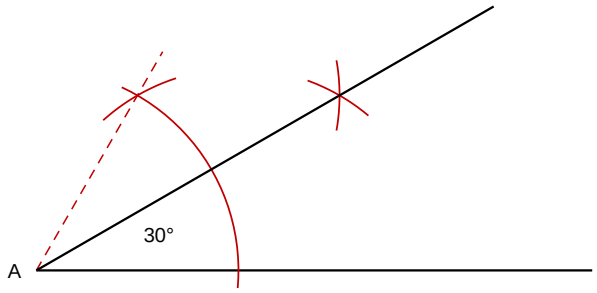
Unit 4 · Perpendiculars and a 30° angle (Lesson 4.2)

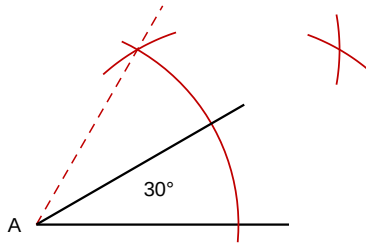
Q	Working	Answer
the perpendicular from a point to a line		

3.1		**Arc from P cuts the line twice; equal arcs from those points meet below the line; join P to that point**
3.2		**Arc from P cuts the line twice; equal arcs from those points meet below the line; join P to that point**
3.3		**Arc from P cuts the line twice; equal arcs from those points meet below the line; join P to that point**
3.4		**Arc from P cuts the line twice; equal arcs from those points meet below the line; join P to that point**

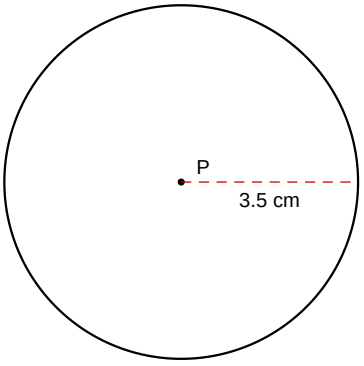
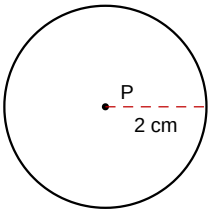
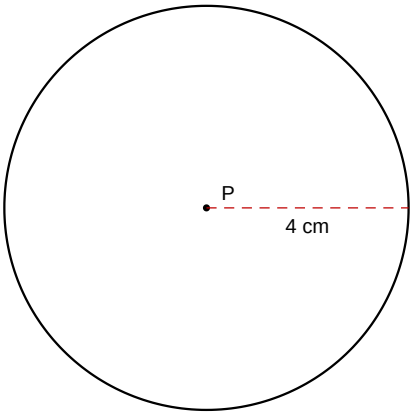
3.5		**Arc from P cuts the line twice; equal arcs from those points meet below the line; join P to that point**
3.6		**Arc from P cuts the line twice; equal arcs from those points meet below the line; join P to that point**
the perpendicular at a point on a line		
3.7		**Equal arcs from A cut the line either side; larger equal arcs from those points meet above A; join**
3.8		**Equal arcs from A cut the line either side; larger equal arcs from those points meet above A; join**

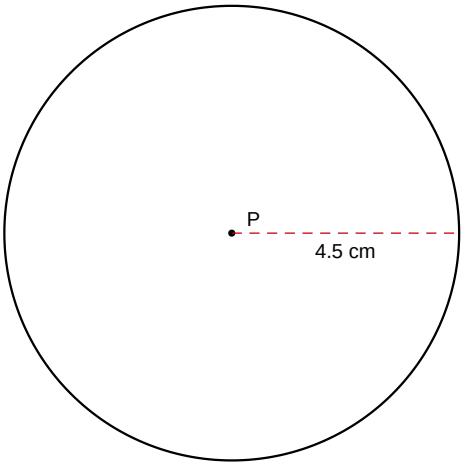
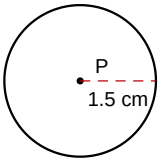
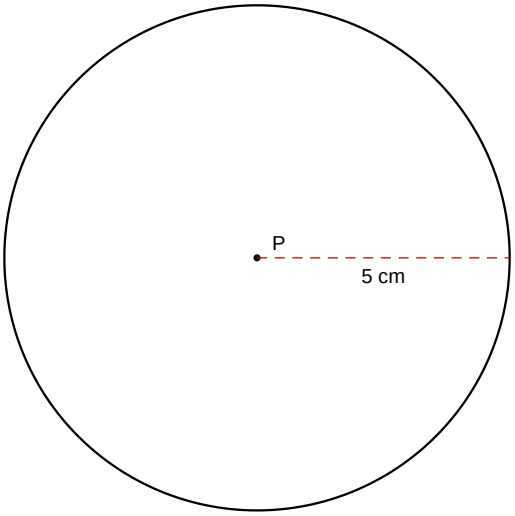
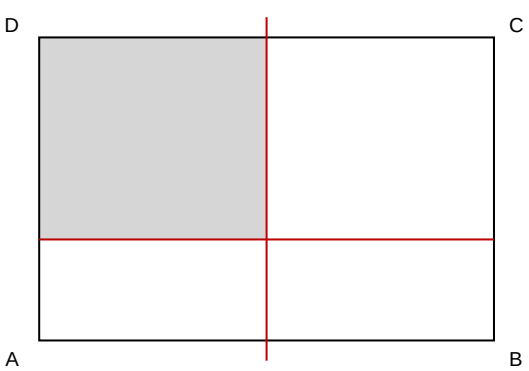
3.9		**Equal arcs from A cut the line either side; larger equal arcs from those points meet above A; join**
3.10		**Equal arcs from A cut the line either side; larger equal arcs from those points meet above A; join**
3.11		**Equal arcs from A cut the line either side; larger equal arcs from those points meet above A; join**
3.12		**Equal arcs from A cut the line either side; larger equal arcs from those points meet above A; join**
an angle of 30°		
3.13		**Construct 60° (equal arcs), then bisect it**

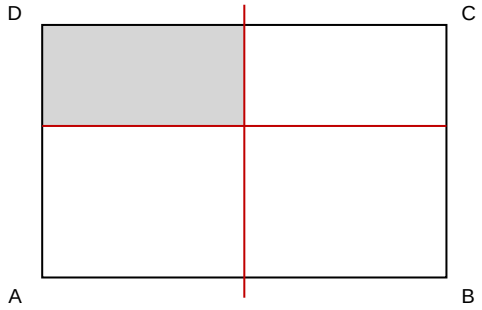
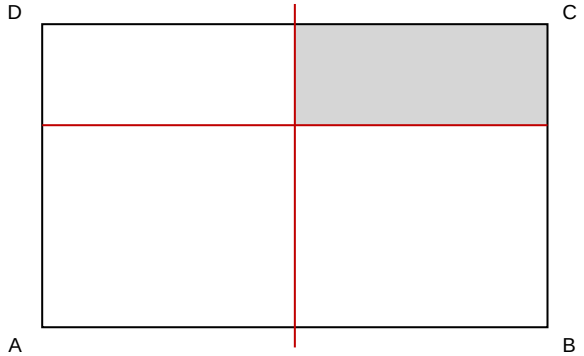
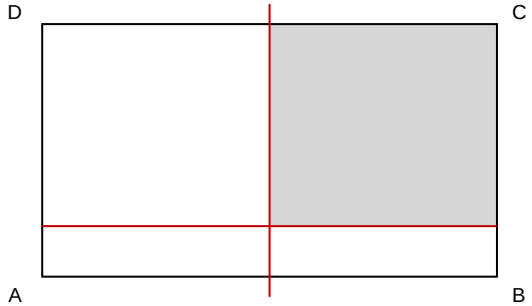
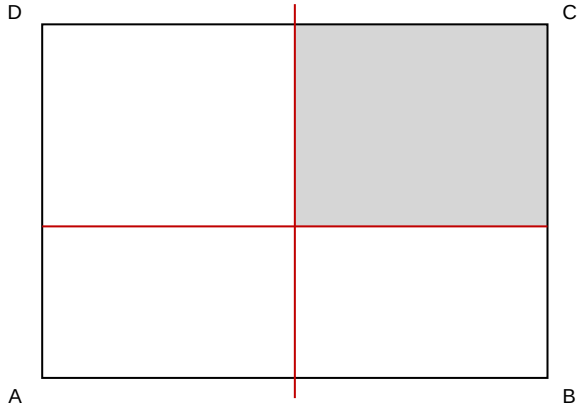
3.14		**Construct 60° (equal arcs), then bisect it**
3.15		**Construct 60° (equal arcs), then bisect it**
3.16		**Construct 60° (equal arcs), then bisect it**
3.17		**Construct 60° (equal arcs), then bisect it**

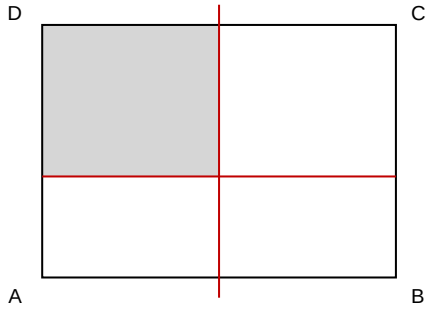
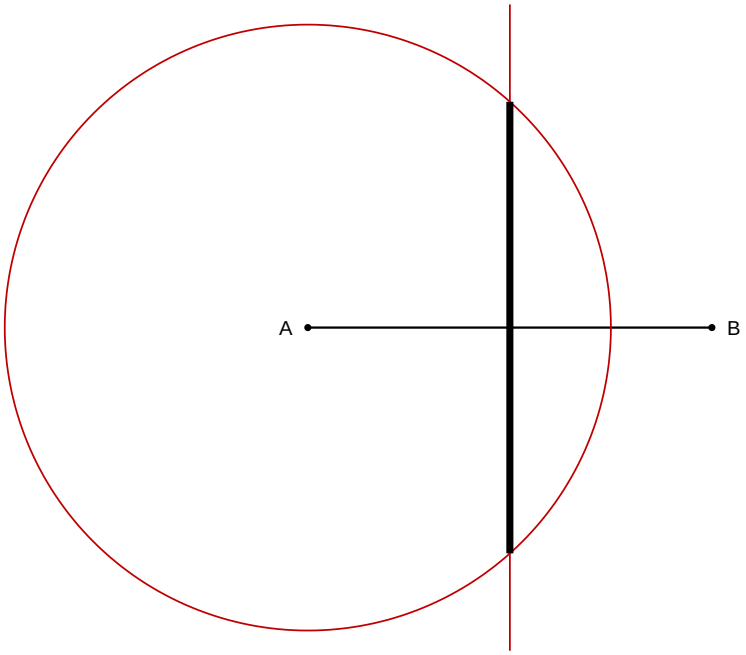
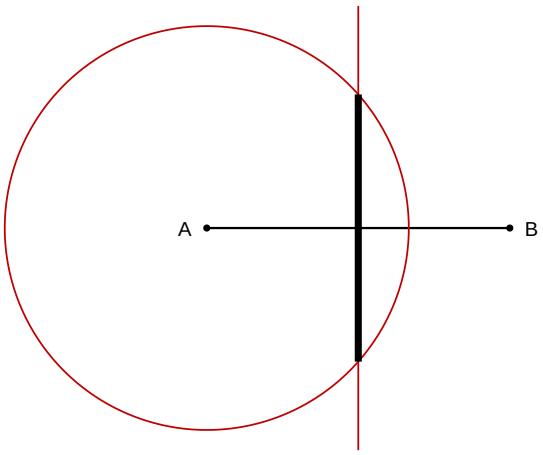
3.18		**Construct 60° (equal arcs), then bisect it**

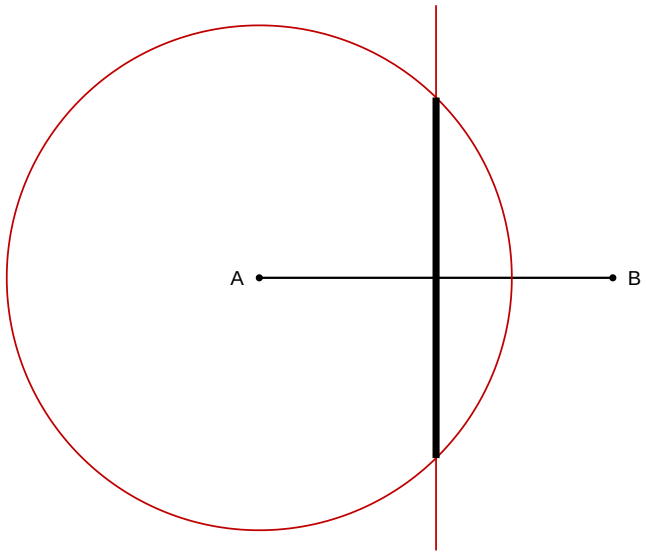
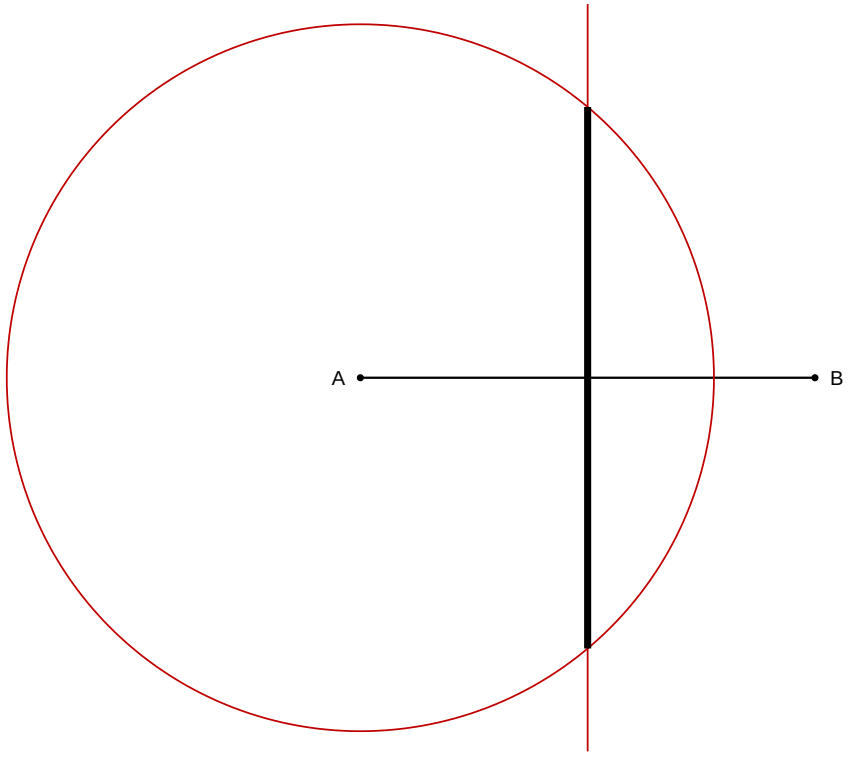
Unit 4 · Loci and regions (Lesson 4.3)

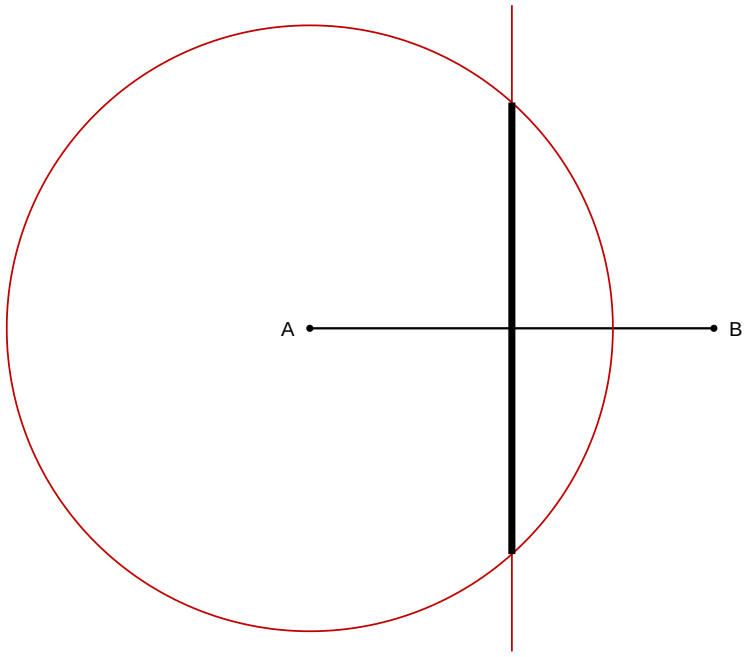
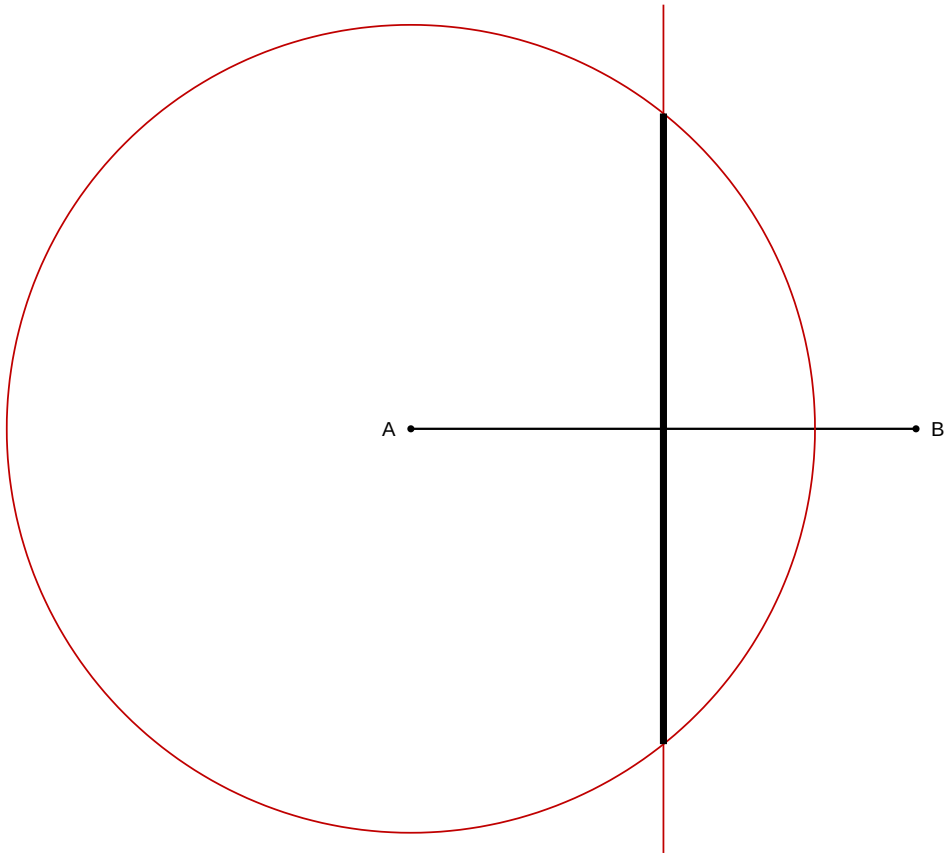
Q	Working	Answer
the locus of points from a point		
4.1		**A circle, centre P, radius 3.5 cm**
4.2		**A circle, centre P, radius 2 cm**
4.3		**A circle, centre P, radius 4 cm**

4.4		**A circle, centre P, radius 4.5 cm**
4.5		**A circle, centre P, radius 1.5 cm**
4.6		**A circle, centre P, radius 5 cm**
regions in a rectangle		
4.7		**Line 2 cm above AB and the perpendicular bisector of DC; shade above the line on D's side**

4.8		**Line 3 cm above AB and the perpendicular bisector of DC; shade above the line on D's side**
4.9		**Line 4 cm above AB and the perpendicular bisector of DC; shade above the line on C's side**
4.10		**Line 1 cm above AB and the perpendicular bisector of DC; shade above the line on C's side**
4.11		**Line 3 cm above AB and the perpendicular bisector of DC; shade above the line on C's side**

4.12		**Line 2 cm above AB and the perpendicular bisector of DC; shade above the line on D's side**
a problem with a scale		
4.13	$80 \div 10 = 8$ cm apart $60 \div 10 = 6$ cm 	**Circle radius 6 cm about A; the mast is on the perpendicular bisector of AB inside the circle (bold)**
4.14	$60 \div 10 = 6$ cm apart $40 \div 10 = 4$ cm 	**Circle radius 4 cm about A; the mast is on the perpendicular bisector of AB inside the circle (bold)**

4.15	$140 \div 20 = 7$ cm apart $100 \div 20 = 5$ cm 	**Circle radius 5 cm about A ; the mast is on the perpendicular bisector of AB inside the circle (bold)**
4.16	$90 \div 10 = 9$ cm apart $70 \div 10 = 7$ cm 	**Circle radius 7 cm about A ; the mast is on the perpendicular bisector of AB inside the circle (bold)**

4.17	$40 \div 5 = 8$ cm apart $30 \div 5 = 6$ cm 	**Circle radius 6 cm about A ; the mast is on the perpendicular bisector of AB inside the circle (bold)**
4.18	$100 \div 10 = 10$ cm apart $80 \div 10 = 8$ cm 	**Circle radius 8 cm about A ; the mast is on the perpendicular bisector of AB inside the circle (bold)**

map scales

4.19	**(a)** $8.5 \times 2 = 17$ **(b)** $13 \div 2 = 6.5$	**(a) 17 km** **(b) 6.5 cm**
4.20	**(a)** $3.4 \times 25 = 85$ **(b)** $105 \div 25 = 4.2$	**(a) 85 km** **(b) 4.2 cm**
4.21	**(a)** $4.7 \times 10 = 47$ **(b)** $62 \div 10 = 6.2$	**(a) 47 km** **(b) 6.2 cm**
4.22	**(a)** $9.2 \times 3 = 27.6$ **(b)** $27 \div 3 = 9$	**(a) 27.6 km** **(b) 9 cm**
4.23	**(a)** $5.5 \times 20 = 110$ **(b)** $130 \div 20 = 6.5$	**(a) 110 km** **(b) 6.5 cm**
4.24	**(a)** $3.5 \times 8 = 28$ **(b)** $52 \div 8 = 6.5$	**(a) 28 km** **(b) 6.5 cm**

Unit 4 · Can the triangle be constructed? (Lesson 4.1)

Q	Working	Answer
can the triangle be constructed?		
5.1	$5 + 6 = 11,\ 11 > 8$	**Yes**
5.2	$2 + 5 = 7,\ 7 < 10$	**No**
5.3	$4 + 4 = 8,\ 8 < 9$	**No**
5.4	$6 + 7 = 13,\ 13 > 12$	**Yes**
5.5	$3 + 5 = 8$, the same as the longest side The sides would lie flat on a straight line	**No**
5.6	$7 + 9 = 16,\ 16 > 15$	**Yes**